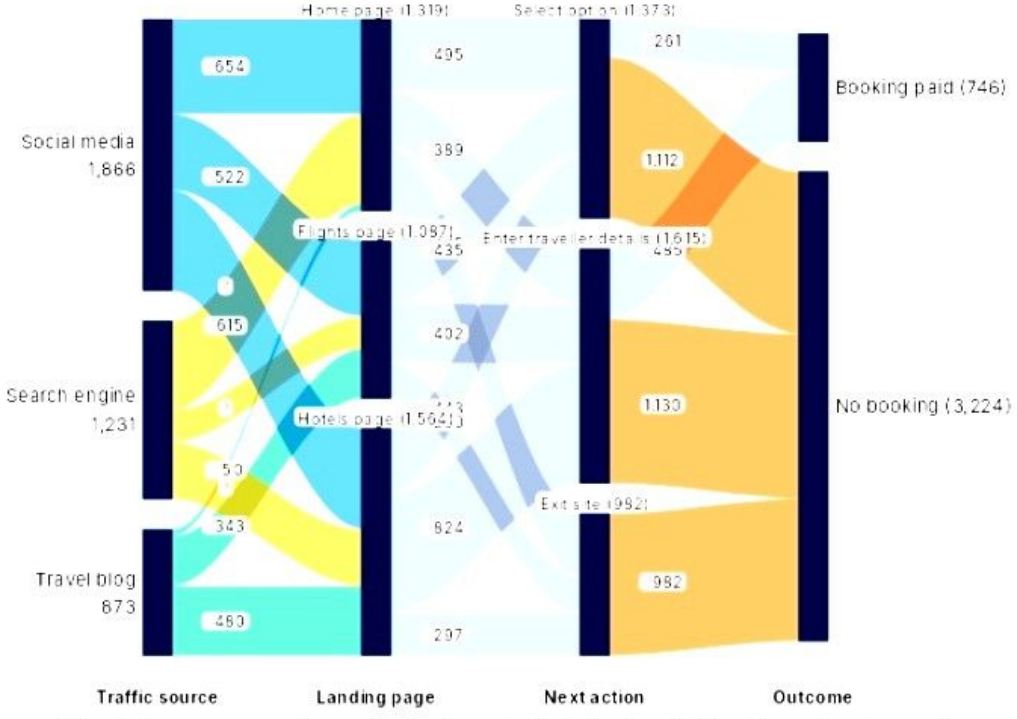
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot C

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Scenario: The analytics team of a travel-booking website tracked 3,970 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Social media to Hotels page; Search engine to Flights page; Search engine to Hotels page. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Booking paid as a percentage of all sessions; (ii) the largest single flow into "No booking" or "Exit site"; (iii) the percentage of "Flights page" sessions that go on to "Enter traveller details". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 200 sessions. Compute the height in cm of each landing-page node (Hotels page, Flights page, Home page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Search engine" and "Travel blog" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

Activity 6:
Website flow Sankey Design

(a) ~~Find the three~~
Hidden flows

Rule: Total incoming flow = Total outgoing flow.

1. Social media to Hostels: 1866 - 654 - 522 - 615 = 75 sessions
2. Search engine to flights : 1231 - 615 - 50 - 343 = 223 sessi
3. Search engine to Hostels : 343 sessions

(b) Calculations

1. Conversion rate. $\frac{746}{3970} \times 100 = 18.79\%$
2. Largest flow into No booking : 1,130 sessions
3. Flights to Enter traveller details : $\frac{435}{1087} \times 100 = 40.02\%$

(c) Four design decisions

1. Node order: Arrange stages from left to right..
2. Colours : Use different colours for diff flows.
3. Labels : shows node names & session counts.
4. Highlighting: Highlight Booking paid to show successfu conversions.

(d) Node heights

Formula : Height = Sessions ÷ 200

- Hostels page : 1564/200 = 7.82cm
- Flights page : 1087/200 = 5.44 cm
- Home page : 1319/200 = 6.60 cm

(e) R code

~~install. packages ("network~~

(e) R code

```
install.packages("networkD3")
library(networkD3)

nodes <- data.frame(name = c(
"Social media", "Search engine", "Travel blog", "Home page",
"Flights page", "Hotels page"))

Links <- data.frame(
Source = c(0,0,0,1,1,1,2,2,2),
target = c(3,4,5,3,4,5,3,4,5),
value = c(654, 522, 690, 615, 222, 394, 50, 343, 480))

Sankey Network(
Links = Links, Nodes = nodes,
Source = "source", Target = "target",
value = "value", NodeID = "name"
```

(f) Two limitations

1. Too many flows cause clutter, use a bar chart
2. Small values are difficult to compare; use a table